

A Practical Guide to Ranking Data Analysis in the Social Sciences

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Examples in Political Sciences

Study	Area	Predictor	Ranking Outcome
Tate (2003)	AP	individual race	types of rep.
McMurry (2022)	CP	priming	sources of identity
Dietrich (2016)	IR	donor origin	foreign aid channels
Atsusaka (2025)	Method	ballot order	candidate rankings

Practical Ranking Data Analysis

- Statistical computing
 - **rdy**: <https://github.com/YukiAtsusaka/rdy>
 - **rankingQ**: <https://sysilviakim.com/rankingQ>

1. Data
collection

2.
Descriptive
statistics

3. Rank
distances

4. Modeling

5. Drawing
quantities of
interest

6.
Sampling/
simulations

7. Power
calculation

8. Missing
data

9.
Extensions

Motivating Application

Here is a list of activities that some citizens find important when thinking about their member of Congress in Washington, DC. Please rank each of these activities in terms of how important it is to you when thinking about your own representative. Rank each item with a number between 1 and 5.

- **policy:** performing in Congress on bills concerning national issues
- **service:** helping people in the district who have personal problems with government
- **allocation:** making sure the state/district gets its fair share of government money and projects
- **contact:** keeping in touch with the people about what the government is doing
- **trust:** making public gestures, appearances, and speeches to gain the trust and confidence of citizens

↪ Black and white Americans have different ranked preferences for representation ([Atsusaka & Hayes, 2025](#))

Motivating Application

	ranking	policy	service	allocation	contact	trust	race
1	24351	2	4	3	5	1	1
2	14235	1	4	2	3	5	1
3	25413	2	5	4	1	3	1
4	21435	2	1	4	3	5	3
5	15324	1	5	3	2	4	1
6	25314	2	5	3	1	4	1

1. Collecting Data

Practical considerations

- Clarifying purposes
 - Eliciting preferences
 - Mimicking real-world decision-making
 - ↪ Num of items (ideally < 7)
 - ↪ Ranking format (drag-and-drop, matrix)
- Measurement error
 - Serious bias due to random responses (Atsushaka and Kim, 2024)
 - Design-based solution
 1. **Randomize** item order
 2. Add an anchor question, with the **correct ranking**
 - ↪ Bias correction (plug-in/inverse probability weighting)

1. Collecting Data

Anchor ranking question

In American politics, there are multiple levels of government, where elected officials can represent their constituents. Please rank the following five levels of government from the highest to the lowest. The level of government is higher when it covers a broader range of government.

- **Federal government**
- **State government**
- **County government**
- **Municipal government**
- **School board**

Key idea: $p(\text{non-random} \mid \text{main}) = p(\text{correct} \mid \text{anchor}) - \frac{1}{5}$ (≈ 0.64)

2. Descriptive Inference

- Key goals
 - Understand overall patterns
 - Show baseline rankings (control units)
 - ↪ **Lots of distinct values:** $J!$ full rankings for J items (e.g., 120 rankings for five items)
- Many ways to “squash” rankings as vectors into scalars
 - **Modal** ranking
 - **Empirical distribution** of ranking profiles
 - Average ranks
 - Pairwise ranking probability
 - top-k ranking probability
 - Marginal ranking probability
 - ↪ A lot of quantities of interest!

2. Descriptive Inference

(Equivalent of univariate analysis)

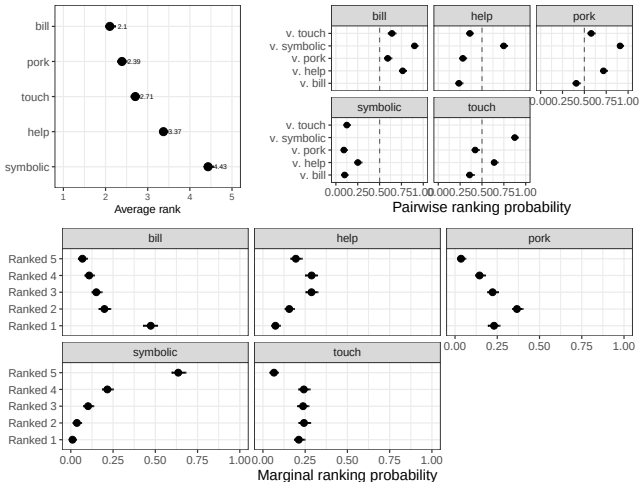
```
rankingQ::impr_r_direct(data = rep, "c_rep", "anchor")
```

ranking	ordering	%
13245	policy > allocation > service > contact > trust	5.4
14235	policy > allocation > contact > service > trust	5.4
14325	policy > contact > allocation > service > trust	4.2
15234	policy > allocation > service > trust > service	3.4

Table: Modal Rankings among Americans

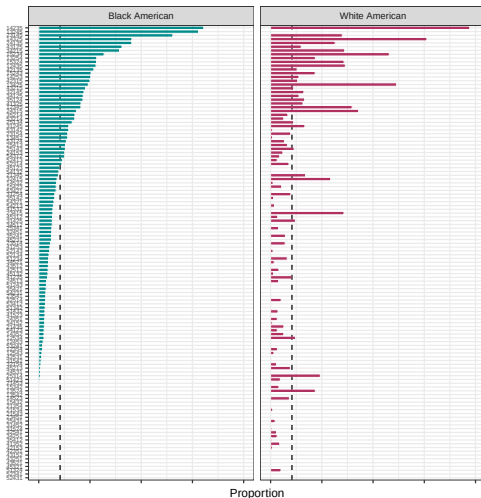
2. Descriptive Inference (Equivalent of univariate analysis)

```
rankingQ::imprr_direct(data = rep, "c_rep", "anchor")
```



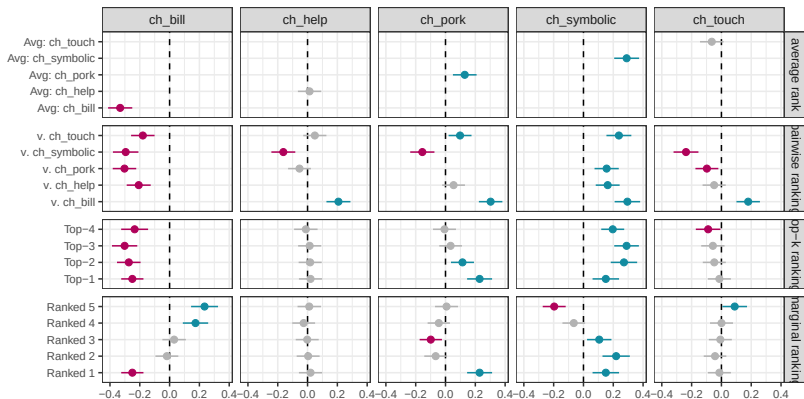
3. Quantifying Differences (Equivalent of bivariate analysis)

```
rankingQ::impr_r_weight(data = rep, "c_rep", "anchor")
```



3. Quantifying Differences (Equivalent of bivariate analysis)

```
rdy::diff_in_mean(dt, item_names, "black") %>%  
  rdy::viz_diff(scale = "std")
```



3. Quantifying Differences (Equivalent of bivariate analysis)

```
rdy::rank_dist(dt, group_col = "black",  
rank_cols = "items", method = "kendall", se = "bootstrap")
```

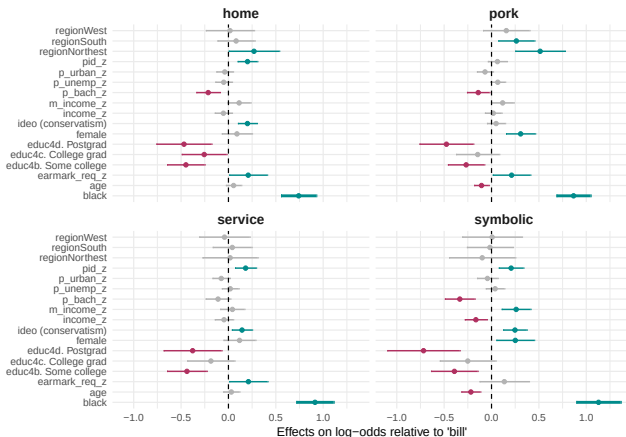
Method	$d_{\max}$	$\hat{\theta}$ (0–1)	SE	95% CI
Kendall τ (# pairwise disagreement)	10	0.45	0.003	[0.44, 0.45]
Footrule (sum of absolute difference)	12	0.61	0.004	[0.60, 0.61]
Spearman's ρ (correlation)	40	0.43	0.004	[0.43, 0.44]

Table: Difference between Rankings by Black and White Americans over Five Items ($J = 5$, $n_W = 1388$, $n_B = 1201$).

4. Modeling via the Plackett-Luce Model

(Equivalent of multivariate analysis)

```
mlogit::mlogit(ch ~ 1 | female + age + educ + ideology +  
pid + income + region, data = mdat, weight = dual_weight,  
reflevel = "policy")
```



5. Drawing Key Quantities of Interest

“clarify”-style simulation-based inference

- Change values of the target independent variable
- Fix all other values to “typical values”
- Draw 500 rankings from Plackett-Luce with the target variable, typical values, and estimated coefficients
- Compute average ranks (or any other functions) over the values of the target variable
- Repeat these steps 1000 times to get 95% CI

→ rdy-function forthcoming

6. Sampling and Simulations

With single means

- Draw from a Plackett-Luce distribution with a single probability vector

```
rdy::rpluce(n = 20, t = 4,  
prob = c(0.4, 0.3, 0.2, 0.1),  
choices = c("safe", "perform", "revenue", "biling"))
```

	1st	2nd	3rd	4th
1	safe	perform	biling	revenue
2	perform	revenue	biling	safe
3	biling	revenue	safe	perform
4	safe	perform	revenue	biling
5	safe	biling	perform	revenue
6	perform	safe	biling	revenue

6. Sampling and Simulations

With conditional means (multiple groups)

- Instead of using a common probability vector
- Define a unit-specific probability vector (i.e., conditional mean)
- Draw from i.i.d Plackett-Luce distributions

id	ch_safe	ch_perform	ch_revenue	ch_biling	age	kids
1	1	4	2	3	36	0
2	2	4	1	3	38	0
3	1	4	2	3	25	1
4	2	4	1	3	45	1
5	1	4	2	3	26	1
6	1	4	2	3	34	1

6. Sampling and Simulations

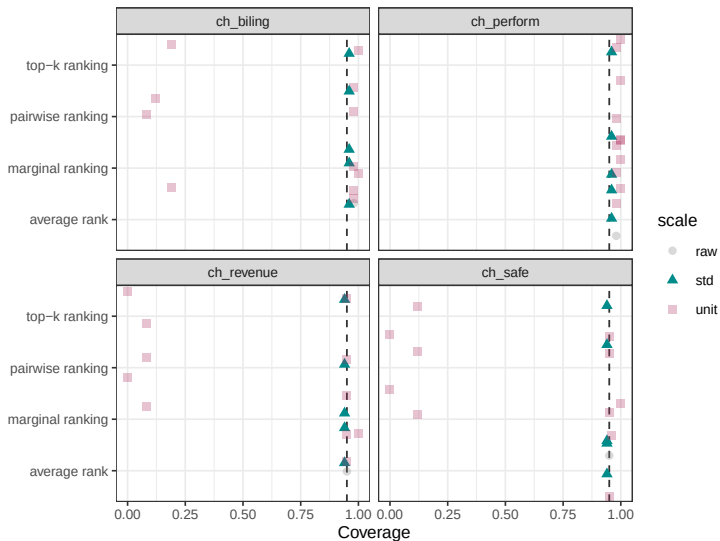
With treatment assignment

- Generate potential rankings under the control (as before)
- Define the **individual** causal effect on **each item's utility**, generate potential rankings under the treatment
- Define treatment assignment (e.g., $D = \text{rbinom}(n, \text{size} = 1, \text{prob} = 0.5)$)
- Generate observed outcomes using $Y_{\text{obs}} = Y(1)D + Y(0)(1-D)$
- Randomly **sample** from the population of observed outcomes

id	treat	ch_safe	ch_perform	ch_revenue	ch_biling	age	kids
1	0	1	4	2	3	36	0
2	1	2	4	1	3	38	0
3	0	1	4	2	3	25	1
4	1	2	4	1	3	45	1
5	1	1	4	2	3	26	1
6	1	1	4	2	3	34	1

7. Inference and Power Calculation

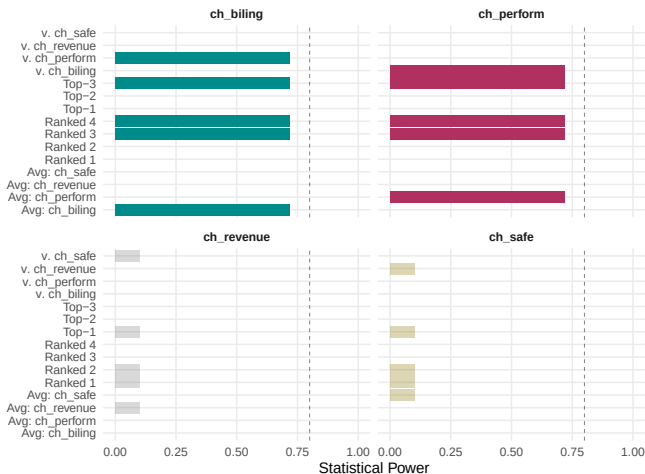
Robust inference with standardized differences ($\tau_{\text{ATE}}/sd(Y_i|D_i = 0)$)



7. Inference and Power Calculation

Simulation-based power analysis

```
rdy::sim_power(pop, n_sample = 500, n_rep = 50)
```



7. Inference and Power Calculation

The Problem of Degenerated Outcomes in the Control

- All control units have the same preference with respect to a given item
 - $\text{Var}(Y_{\text{obs}}|D=0) = 0$
 - No information about the distribution (effect size is unknown)
 - Standardized diff-in-means will be undefined
- What degeneracy may imply
 - No heterogeneity in the preference among controls
 - The sample is too small to spot variation
 - Potential floor/ceiling effects

8. Missing Data

- Partially ranked data
 - One or more items are not ranked
 - Behavioral, design, both
 - ↪ Again, clarify purposes when collecting data
- Statistical issues
 - Listwise deletion induces bias **even under MCAR**
 - Existing methods are not appropriate in social science
 - ↪ Non-parametric bias correction ([Atsusaka and Liu, 2025](#))

Level	Variable	Missing
Candidate	Peltola	0.41
Candidate	Bye	0.64
Candidate	Begich	0.35
Candidate	Palin	0.43
Rank	rank1	0.00
Rank	rank2	0.33
Rank	rank3	0.67
Rank	rank4	0.81

References

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